ChatGPT

On-Device Meeting Assistant Hackathon: Competitor & Feasibility Analysis

Executive Summary

We surveyed existing smartphone and app solutions for meeting transcription and action-item extraction. Current phone-native AI features (e.g. Google Pixel Recorder, Samsung Galaxy AI, OriginOS AI Recorder) focus on **on-device transcription and summaries**, but none extract task assignments with deadlines or check calendar conflicts. Third-party apps (Loqui, Otter.ai, VibeMeeting, etc.) do extract action items and decisions【87†L1-L4】【107†L55-L64】, but rely on cloud processing and do not bridge to a laptop. No known product implements the full pipeline (on-device listening → commitment extraction → conflict detection → OfficeKit handoff). In summary, the pieces exist in isolation but *not end-to-end*. Our hackathon can uniquely combine local speech recognition, custom NLP for tasks, calendar logic, and OriginOS Office Kit integration to fill this gap.

Competitor Matrix

Product (Platform)	On-Device ASR?	On-Device LLM?	Commitment Extraction?	Calendar Conflict Detection?	Phone→PC Handoff?	Phone-First / NPU Use?	Priv
Samsung Galaxy AI (One UI)	Yes (live call transcription) 【103†L276-L285】	Yes (local summaries) 【103†L276-L285】	Yes – detects action items and can "Send to Tasks"【103†L276-L285】	No (no known calendar conflict feature)	Partial (Samsung DeX desktop share, no structured tasks)	Yes (uses phone NPU)	On- (no cor
OriginOS / iQOO AI Recorder	Yes (real-time meeting transcription) 【47†L7-L10】	Yes (AI summaries) 【47†L7-L10】	No (only summary, no explicit task parsing)	No	Yes – Office Kit links phone and PC 【94†L616-L624】	Yes (phone-first, NPU acceleration)	Off opt
Google Pixel Recorder (Gemini Nano)	Yes (on-device, offline) 【78†L63-L68】	Yes (on-device Gemini summary) 【78†L63-L68】	No (generates brief summaries but no structured tasks)	No	No	Yes (phone NPU for ML)	Full (au dev

Product (Platform)	On-Device ASR?	On-Device LLM?	Commitment Extraction?	Calendar Conflict Detection?	Phone→PC Handoff?	Phone-First / NPU Use?	Priv
Loqui (mobile app)	No (cloud via AssemblyAI) 【87†L1-L4】	No (cloud via Anthropic Claude) 【87†L1-L4】	Yes – extracts to-do action items 【87†L1-L4】	No	No	Yes (designed for mobile)	Cla aft tra L4】
Hyprnote (app)	Yes (local Whisper/ Silero) 【60†L733-L740】	Yes (local summaries via LLM or API) 【60†L733-L740】	Partial (creates summaries, but action-item support unclear)	No	No	Yes (fully on-device)	“10 tra not ser L74
Jamie (app)	No (needs cloud)	Yes (cloud AI)	Yes – transcripts, summaries, action items【58†L639-L644】	No	No	Partially (mobile client)	EU/ del use
Otter.ai	No (cloud service)	Yes (cloud AI)	Yes – real-time transcription with action items 【60†L783-L792】	No	No	Partially (mobile app)	GD has 【6
VibeMeeting (iOS)	Yes (offline AI)	Yes (offline LLM)	Yes – topics, decisions, action items, tasks 【107†L55-L64】	No	No	Yes (on-device models)	Dat by upl L64

Each cell cites relevant sources. For example, Samsung’s One UI 8 feature “Real-Time Call Assist” transcribes calls on-device and identifies action items【103†L276-L285】, but neither it nor Pixel’s Recorder checks calendar conflicts or uses a PC bridge. Loqui and Otter can extract tasks but rely on cloud AI【87†L1-L4】【60†L783-L792】. VibeMeeting is close on-device (local transcription and tasks) 【107†L55-L64】, but no conflict-check and iOS-only.

Product Features & Notes

- **Samsung Galaxy AI / Voice Recorder (One UI 8+)** – The built-in recorder and assistant can *live-transcribe calls and meetings*, distinguish speakers, and even auto-generate summaries and “follow-up” action items that can be sent to the Samsung Tasks app【103†L276-L285】. This uses on-device ML (Galaxy NPU) for speech-to-text and simple intent tagging. It is highly privacy-oriented (all processing is local). *Missing:* no automatic deadline extraction or calendar conflict checking, and no PC handoff beyond general Samsung DeX features.

- **OriginOS (vivo/iQOO) AI Recorder** – Vivo’s OriginOS 6 introduced an AI Meeting Assistant that records meetings (on-device) and produces structured minutes with speaker roles and summaries【47†L7-L10】. It supports various templates (e.g. “meeting summary”, “lecture notes”). OriginOS also includes an *Office Kit* feature for seamless phone–PC collaboration

【94†L616-L624】. *Limitations:* It does not explicitly extract "who will do what by when", nor check for conflicts. Workflow integration to PC is generic (file sharing), not structured task export.

- **Google Pixel Recorder (Gemini)** – On Pixel phones, the Recorder app transcribes audio fully offline and generates brief summaries with the Gemini Nano LLM【78†L63-L68】. It is extremely privacy-friendly (everything on-device). However, it does **not** identify action items or deadlines, and has no integration to other apps like calendar or PC. It is phone-first with no cloud component.

- **Loqui (third-party app)** – Loqui is a mobile app that records meetings and *uses* cloud AI: audio is sent to AssemblyAI for transcription and to Anthropic's Claude for summary/action-item extraction【87†L1-L4】. It then *deletes* the raw audio from the device and syncs transcripts to Google Drive. Features include speaker separation, bullet summaries, and to-do lists, but **all processing is in the cloud**. There is no on-device mode, nor any calendar or PC integration. Privacy claim: audio is deleted after use【87†L1-L4】.

- **Hyprnote (app)** – Hyprnote is a local-first note-taking app that records meetings and summarizes them *locally on the device*【60†L733-L740】. It uses Whisper-style ASR and on-device LLMs (with optional custom API keys) to produce transcripts and general summaries. Data never leaves the phone by default. It supports offline use and integration with local notes. It does not specifically target "commitments" — it provides free-form summaries — and has no built-in calendar or PC sync.

- **Jamie (meeting assistant)** – Jamie is a well-known "AI note-taker" that passively listens to meetings (bot-free) and produces transcripts, summaries, and action items【58†L639-L644】. It strongly emphasizes privacy (EU-hosted, GDPR-compliant) and deletes audio after transcription. However, it is **not on-device** — audio is recorded and processed in the cloud. Jamie also does not connect directly to calendars or PCs.

- **Otter.ai** – Otter's AI Agent joins online meetings and provides real-time transcripts, summaries, and highlights (including identified action items)【60†L783-L792】. It integrates with calendars and collaboration tools to pull in meeting audio automatically. Like Jamie, Otter is primarily cloud-based (and was not designed for offline meeting capture). There is no on-device processing or direct phone-to-PC handoff.

- **VibeMeeting (iOS)** – VibeMeeting is a local-first app for iPhone that transcribes meetings on-device (using local AI models) and generates *structured meeting records*, including topics, decisions, risks, and action items【107†L55-L64】. It stores transcripts and audio locally for privacy, with optional cloud API for translations. This is the closest existing tool: it **extracts action items and tasks on-device** and keeps data private. *Gaps:* It still lacks deadline inference and calendar conflict checks, and it is iOS-only with no PC bridging feature.

Gap Analysis

No single product covers the full workflow. For example, Samsung and OriginOS do on-device transcribe and even detect tasks, but **do not tie tasks to dates or calendars**, nor send them to a laptop app. Pixel's Recorder is on-device and private, but it only summarizes, not extract tasks. Loqui/VibeMeeting extract tasks but rely on the cloud (Loqui) or are platform-limited. **Calendar conflict detection is absent** in all: none of the above checks an existing schedule to flag overlaps. Likewise, **no product**

outputs a structured task list to a PC app via something like Office Kit. Thus each candidate misses at least one key piece (see table). We conclude that our proposed pipeline – on-device ASR + commitment parsing + calendar logic + OfficeKit handoff – would be *unique*.

Technical Feasibility Notes

Implementing this pipeline on a Snapdragon phone is challenging but doable with today's tech. On-device ASR can use lightweight models (e.g. Whisper tiny/SMALL or alternatives like Silero/Vosk) optimized via NNAPI on the NPU【104†L248-L257】. A Whisper-small model (~70M params) can run in roughly real-time on modern NPUs; even Whisper-large can be quantized or segmented if needed. For commitment extraction, a small custom NLP model or rule-based parser can identify patterns like "*<Person> will <verb> by <date/time>*". A compact text model (e.g. DistilBERT or a small LLM) could classify sentences or extract date entities. Deadline/time extraction can use known NER models. Speaker diarization or overlap handling is hard on-device; we may skip overlap or assume a single note-taker. All processing must fit within a ~2-3 GB memory and run under a few seconds latency; we likely limit summaries to a few key lines. Confidence scores can be derived from ASR token probabilities or model output (e.g. Speech-to-text often gives per-word confidence). Calendar conflict detection is straightforward via the Android Calendar API.

Privacy advantage: "audio never leaves the phone" means strong security【104†L239-L241】. We must ship ASR and NLP libs as Android libraries (TensorFlow Lite or ONNX Runtime for mobile). Office Kit integration (OriginOS's phone–PC bridge) can send structured JSON (task lists with fields) rather than audio. The development risk: accuracy on multi-speaker, noisy real meetings is tough, and on-device LLM summarization is slower/less accurate than cloud. We can mitigate by focusing on clear highlights and using confidence thresholds to avoid false tasks.

Recommendations & MVP Plan

- **Unique Angles (Hackathon Differentiators):**
- **Fully offline and private pipeline:** Emphasize that *nothing* audio-related leaves the phone, unlike cloud tools【104†L239-L241】.
- **Integrated calendar conflict alerts:** If a extracted deadline conflicts with an existing appointment, highlight it in real time (no other app does this).
- **Speaker-aware commitments:** Attributed tasks ("[Name] will ...") with speaker labels. Possibly snap photo of meeting invite to recognize attendees vs. transcript voices.
- **Contextual push to PC:** Use Office Kit to seamlessly push the *structured task list* (with fields: owner, action, due date, confidence) to a laptop app window, not raw notes.
- **Confidence tagging:** Show an AI confidence score for each action item, so users know which items may need manual verification.
- **Minimal MVP Scope (30 hours):** Build a prototype on an iQOO/OriginOS phone using available tools (e.g. Whisper-CPP for ASR, a small rule-based extractor, Android Calendar API, and OriginOS Office Kit interface). Focus on just **record → detect tasks → show on phone → send to PC**. Skip advanced LLM summarization or UI polish.
- **Mermaid Timeline (30-hour plan):**

```
gantt
    title 30-Hour Hackathon Schedule
    dateFormat  H:mm
    axisFormat  %H:%M
    section Setup
    Environment & Tools         :done, 0:00, 2h
    section Transcription
    Integrate on-device ASR     : 2:00, 4h
    Test & Optimize ASR         : 6:00, 2h
    section Task Extraction
    Develop NLP extractor       : 8:00, 4h
    Calendar conflict logic     : 12:00, 2h
    section Integration
    Implement OfficeKit bridge  : 14:00, 2h
    Combine pipeline & test     : 16:00, 4h
    section Finalization
    UI polish & bug fixes       : 20:00, 4h
    Demo script & rehearsal     : 24:00, 6h
```

- **Architecture Diagram:**

```
flowchart LR
    Mic[Phone Microphone] --> ASR[On-Device ASR Model (NPU)]
    ASR --> NLP[Commitment Extractor / LLM]
    NLP --> Tasks{Extracted Tasks List}
    Tasks --> +Cal[Read Calendar Events]
    Tasks --> Output[Structured Task JSON]
    +Cal --> Conflicts[Conflict Detector]
    Tasks -->|with conflicts| Output
    Output --> OfficeKit[Phone→PC Bridge]
    OfficeKit --> PC[Laptop App (Tasks List)]
```

This sketch shows the phone's microphone feeding an ASR model; the text flows into a commitment-extraction module (identifying owner, action, deadline), then cross-references the calendar for conflicts. The final structured list is sent via Office Kit to the PC app.

Sources: We drew on official feature docs (Samsung【103†L276-L285】, OriginOS【47†L7-L10】【94†L616-L624】, Google/VOCAP blog【78†L63-L68】) and app descriptions (Loqui【87†L1-L4】, VibeMeeting【107†L55-L64】) for the matrix and feature details. These confirm that while many tools transcribe and even list tasks, none implement the full pipeline end-to-end as we propose.
